2. Carbohydrate and protein solutions are highly viscous in nature. The lubricating property of the synovial fluid is achieved mainly by the viscous nature of the mucopolysaccharides present in the synovial fluid.
3. Dipalmitoyl lecithin is a surfactant that is secreted by the lung alveoli, which reduces the surface tension and prevents the collapse of lung alveoli during expiration. Certain premature infants have low levels of this surfactant leading to acute respiratory distress.

1.5. OSMOSIS

Osmosis is the net movement of water across a semi-permeable membrane from an area of lower concentration of the solute to an area of higher concentration of the solute.

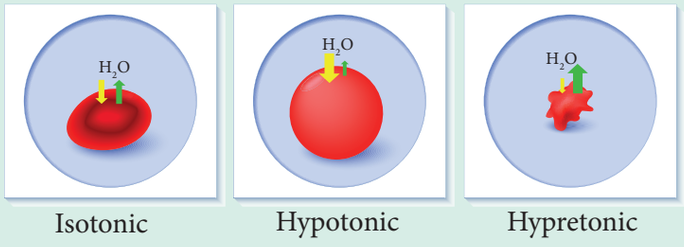


Figure 8: Osmolarity and the consequent changes in a cell

Tonicity is responsible for the movement of the water into or out of the cell. A solution's tonicity is related to its osmolarity, which is the total concentration of all solutes in the solution. A solution with low osmolarity has fewer solute particles per liter of solution, while a solution with high osmolarity has more solute particles per liter of solution. When solutions of different osmolarities are separated by a membrane permeable to water, but not to solute, water will move from the side with lower osmolarity to the side with higher osmolarity.

- If the extracellular fluid has lower osmolarity than the fluid inside the cell, it is said to be hypotonic—hypo means less than—to the cell, and the net flow of water will be into the cell.
- In the reverse case, if the extracellular fluid has a higher osmolarity than the cell's cytoplasm, it is said to be hypertonic—hyper means greater than—to the cell, and water will move out of the cell to the region of higher solute concentration.
- In an isotonic solution—iso means the same—the extracellular fluid has the same osmolarity as the cell, and there will be no net movement of water into or out of the cell.

Hypotonic, hypertonic, and isotonic are relative terms. That is, they describe how one solution compares to another in terms of osmolarity. For instance, if the fluid inside a cell has a higher osmolarity, than the surrounding fluid, the cell interior is hypertonic to the surrounding fluid, and the surrounding fluid is hypotonic to the cell interior.

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